

① Improper Integrals:-

An integral where either the interval of integration is infinite or the integrand becomes infinite (discontinuous) at one or more points in the interval.

Types of Improper Integrals-

1. Infinite limits of Integration [when interval of integration is ∞]

Form: $\int_a^{\infty} \frac{1}{x^r} dx$

Ex: convergent if $r > 1$

Result: Divergent if $r \leq 1$

2. Infinite Discontinuities -

These occur when $f(x)$ becomes infinite at one or more points in $[a, b]$

Form: $\int_a^b \frac{1}{x^p} dx$

Ex: convergent if $p < 1$

Result: Divergent if $p \geq 1$

3. Combination (Infinite interval and Discontinuity) -
when both occur together

Form: $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$

Ex: Always
convergent

Result: $= \pi$

Convergence And Divergence:

- An improper integral converges if the defining limit exists and is finite.
- It diverges if the limit does not exist or is infinite.

$$\textcircled{2} \int_1^{\infty} \frac{\sqrt{x}}{\ln x} dx$$

As $x \rightarrow \infty$ the integrand behaves like $x^{1/2} / \ln x$

For any $\varepsilon > 0$ we have $\ln x \leq x^\varepsilon$ for sufficiently large x . choose $\varepsilon = 1/4$. for x large enough

$$\frac{\sqrt{x}}{\ln x} \geq \frac{\sqrt{x}}{x^{1/4}} = x^{1/4}$$

since $\int_1^{\infty} x^{1/4} dx$ diverges, by comparison the given integral diverges
 $\Rightarrow$ The integral diverges to $+\infty$.

$$\textcircled{3} \int_a^b \frac{dx}{(x-a)^p} \quad (p > 0)$$

$$= \lim_{\varepsilon \rightarrow 0^+} \int_{a+\varepsilon}^b \frac{dx}{(x-a)^p} \quad \begin{array}{l} u = x-a \\ \text{when } x = a+\varepsilon \text{ we have } u = \varepsilon \end{array}$$

$$= \int_{\varepsilon}^{b-a} u^{-p} du$$

CASE A: $p \neq 1$

$$\int_{\varepsilon}^{b-a} u^{-p} du = \left[\frac{u^{1-p}}{1-p} \right]_{\varepsilon}^{b-a} = \frac{(b-a)^{1-p} - \varepsilon^{1-p}}{1-p}$$

Let $\varepsilon \rightarrow 0^+$

$1-p > 0$ (i.e. $p < 1$), then $\varepsilon^{1-p} \rightarrow 0 \Rightarrow$ limit exists & is equal to

$$\int_a^b \frac{dx}{(x-a)^p} = \frac{(b-a)^{1-p}}{1-p} \quad (\text{finite})$$

so the integral converges for $p < 1$ & diverges for $p > 1$

CASE B: $p = 1$

$$\int_{\varepsilon}^{b-a} u^{-1} du = [\ln u]_{\varepsilon}^{b-a} = \ln(b-a) - \ln \varepsilon$$

As $\varepsilon \rightarrow 0^+$, $\ln \varepsilon \rightarrow -\infty$
 so the integral diverges
 (logarithmically).
 divergent for $p = 1$

$$\textcircled{4} \int_{-\infty}^{\infty} \frac{\sin x}{1+x^2} dx$$

$$\int_{-\infty}^{\infty} \left| \frac{\sin x}{1+x^2} \right| dx \leq \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \pi < \infty \quad \text{Hence, integral converges absolutely}$$

$\frac{\sin x}{1+x^2}$ is odd function [because $\sin(-x) = -\sin x$], hence the integral over symmetric $(-\infty, \infty)$ is 0. Convergence absolutely value = 0.

$$\textcircled{5} \int_0^{\infty} \frac{x \tan^{-1} x}{\sqrt{4+x^3}} dx = I$$

$$I = \lim_{R \rightarrow \infty} R \int_0^R \frac{x \tan^{-1} x}{\sqrt{4+x^3}} dx$$

As $x \rightarrow 0^+$, $\tan^{-1} x \sim x$. Also $\sqrt{4+x^3} \rightarrow \sqrt{4} = 2$

$$\Rightarrow \frac{x \tan^{-1} x}{\sqrt{4+x^3}} \approx \frac{x \cdot x}{2} = \frac{x^2}{2}$$

Because, $\int_0^1 x^2 dx$ converges, the integrand is integrable on a neighbourhood 0.

As $x \rightarrow \infty$, $\tan^{-1} x \rightarrow \pi/2$ (a +ve const.) and $\sqrt{4+x^3} \sim x^{3/2}$

$$\frac{x \tan^{-1} x}{\sqrt{4+x^3}} \sim \frac{(\pi/2) x}{x^{3/2}} = \frac{\pi}{2} x^{-1/2}$$

The tail integral $\int x^{-1/2} dx$ diverges

Final conclusion: $\int_0^{\infty} \frac{x \tan^{-1} x}{\sqrt{4+x^3}} dx$ diverges (equals $+\infty$).

$$\textcircled{6} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tan x \, dx$$

$x = \pm \pi/2$. The improper integral is defined only if the two one-sided limits exist.

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tan x \, dx = \lim_{a \rightarrow \pi/2^+} \lim_{b \rightarrow \pi/2^-} \int_a^b \tan x \, dx$$

$$\int_0^{\frac{\pi}{2}} \tan x \, dx = \lim_{b \rightarrow \pi/2^-} (-\ln(\cos b)) = +\infty \quad [\text{for right piece}]$$

$$\int_{-\pi/2}^0 \tan x \, dx = \lim_{a \rightarrow \pi/2^+} (-\ln(\cos 0) + \ln(\cos a)) = \lim_{a \rightarrow \pi/2^+} \ln(\cos a) = -\infty$$

because $\cos a \rightarrow 0^+$ as $a \rightarrow -\pi/2^+$

since two one-sided integrals are not finite

$\int_{-\pi/2}^{\pi/2} \tan x \, dx$ does not converge

Hence, divergent

$$\textcircled{7} \int_0^{\frac{\pi}{2}} \frac{\sin^m x}{x^n} \, dx \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \frac{\cos^m x}{x^n} \, dx$$

• Near 0: $\sin x \sim x$ so $\sin^m x \sim x^m$. The integrand $\sim x^{m-n}$ and for $(\cos)$ near $x = \pi/2$

$$m-n > -1 \Rightarrow n < m+1$$

• for $\cos^m x$ near 0, $\cos 0 = 1$ so no singularity at 0 from cosine is from x^n . The integral $\sim x^{-n}$, $n < 1$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^m x}{x^n} \, dx \text{ converges near 0, if } n < m+1$$

$$\int_0^{\frac{\pi}{2}} \frac{\cos^m x}{x^n} \, dx \text{ converges near 0, if } n < 1$$

① $\int \frac{\sin(1/n)}{n^p} dn$ converges absolutely for $p > 1$

$$\int \frac{|\sin(1/n)|}{n^p} dn \leq \int n^{-p} dn$$

The p integral $\int n^{-p} dn$ converges if $p > 1$, thus the given integral is absolutely convergent for $p > 1$

② $\int_{-\infty}^{\infty} \frac{dn}{n^2 + 5n + 6}$

$$n^2 + 5n + 6 = (n+2)(n+3)$$

$$\Rightarrow n = -2, -3$$

$$= \int_{-\infty}^{\infty} \frac{dn}{(n+2)(n+3)}$$

Partial fraction: $\frac{1}{(n+2)(n+3)} = \frac{1}{(n+2)} - \frac{1}{(n+3)}$

Antiderivative: $f(n) = \ln \left| \frac{n+2}{n+3} \right|$

$$\int_{-\infty}^{\infty} = \int_{-\infty}^{-3} + \int_{-3}^{-2} + \int_{-2}^{\infty}$$

Take the left piece: $\int_{-\infty}^{-3} \frac{dn}{(n+2)(n+3)} = \lim_{A \rightarrow -\infty} [F(-3^-) - F(A)]$

The improper integral diverges because of the poles at $n = -3$ and $n = -2$.
The Cauchy principle value doesn't exist and equals 0.

③ $\int_0^4 \frac{dn}{n^2 - 2n - 8}$

$$n^2 - 2n - 8 = (n-4)(n+2)$$

$$\frac{1}{(n-4)(n+2)} = \frac{1}{6} \frac{1}{(n-4)} - \frac{1}{6} \frac{1}{(n+2)}$$

$$f(n) = \frac{1}{6} \ln \left| \frac{n-4}{n+2} \right|$$

$$\int_0^b \frac{dn}{n^2 - 2n - 8} = f(b) - f(0)$$

$$\frac{1}{(n-4)(n+2)} = \frac{A}{n-4} + \frac{B}{n+2}$$

$$A(n+2) + B(n-4) = 1$$

$$n=4 \quad n=-2$$

$$6A=1 \quad -6B=1$$

$$A=1/6 \quad B=-1/6$$

$$f(0) = \frac{1}{6} \ln \left| \frac{-4}{2} \right| = \frac{1}{6} \ln 2$$

$$b \rightarrow 4^- \Rightarrow n-4 \rightarrow 0^-$$

$$|n-4| \rightarrow 0$$

$$f(b) = \frac{1}{6} \ln \left| \frac{b-4}{b+2} \right| \rightarrow -\infty$$

The integral doesn't converge, it is not finite.

① Gamma function: for $n > 0$

$$\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t} dt$$

at $t=0$
As $t \rightarrow 0^+$ $e^{-t} \approx 1$

$\Rightarrow t^{n-1}$

$\int_0^1 t^{n-1} dt = \frac{1}{n}$, converges if and only if $n > 0$

Thus gamma function converges near 0 if $n > 0$

at $t \rightarrow \infty$
As $t \rightarrow \infty$ e^{-t}

$t^{n-1} e^{-t} \rightarrow 0$ rapidly

Hence for Gamma function, the integral converges at ∞ .

Beta function: for $p, q > 0$ (at least initially)

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt$$

at $t=0$
As $t \rightarrow 0^+$ $(1-t)^{q-1} \approx 1$

Then $\int_0^{\epsilon} t^{p-1} dt = \frac{1}{p} \epsilon^p$, which converges if $p > 0$.

At $t=1$
let $u = 1-t$
 $(1-t)^{q-1} = u^{q-1}$

$\int_{1-\epsilon}^1 (1-t)^{q-1} dt = \int_0^{\epsilon} u^{q-1} du = \frac{1}{q} \epsilon^q$, which converges if $q > 0$

$$(2) \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} t^{-1/2} e^{-t} dt$$

$$t = u^2 \quad dt = 2u du$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} \frac{1}{u} e^{-u^2} (2u du) = 2 \int_0^{\infty} e^{-u^2} du$$

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^{\infty} e^{-u^2} du$$

$$\int_0^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2} \quad (\text{classic Result})$$

$$\Gamma\left(\frac{1}{2}\right) = 2 \times \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$$

$$(3) \Gamma\left(-\frac{5}{2}\right)$$

$$\Gamma(z+1) = z \Gamma(z)$$

$$\text{Rearranging, } \Gamma(z) = \frac{\Gamma(z+1)}{z}$$

$$\text{we know that, } \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right)}{\frac{-1}{2}} = -2\sqrt{\pi}$$

$$\Gamma\left(-\frac{3}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}\right)}{\frac{-3}{2}} = \frac{-2\sqrt{\pi}}{\frac{-3}{2}} = \frac{4}{3}\sqrt{\pi}$$

$$\Gamma\left(-\frac{5}{2}\right) = \frac{\Gamma\left(-\frac{3}{2}\right)}{\frac{-5}{2}} = \frac{\frac{4\sqrt{\pi}}{3}}{\frac{-5}{2}} = -\frac{8}{15}\sqrt{\pi}$$

$$(4) \int_0^{\infty} \frac{x^a}{a^x} dx = \frac{1}{(\log a)^{a+1}} \sqrt{a+1}$$

$$u = (\log a)x \quad \therefore t = \ln a$$

$$I(a) = \int_0^{\infty} \frac{x^a}{a^x} dx = \int_0^{\infty} x^a e^{-x \ln a} dx$$

As $x \rightarrow 0$, $x^a e^{-x \ln a} \sim x^a$. So the integral converges near 0 if $a > -1$

As $x \rightarrow \infty$, the factor $e^{-x \ln a}$ decays only when $\ln a > 0$, $0 < a < 1$

Put $t = x \ln a$

$$x = \frac{t}{\ln a} \quad du = \frac{dt}{\ln a} \quad \text{The limits } x=0 \rightarrow t=0, x \rightarrow \infty, t \rightarrow \infty$$

$$I(a) = \int_0^{\infty} \left(\frac{t}{\ln a} \right)^a e^{-t} \frac{dt}{\ln a} = (\ln a)^{-a-1} \int_0^{\infty} t^a e^{-t} dt$$

The remaining integral is Gamma function $\Gamma(a+1)$, $a > -1$

$$\boxed{\int_0^{\infty} \frac{x^a}{a^x} dx = \frac{\Gamma(a+1)}{(\ln a)^{a+1}}}$$

$$(5) \int_b^1 x^m (\log x)^n dx = \frac{(-1)^n}{(m+1)^{n+1}} \sqrt{n+1}$$

$$t = \log x \quad dx = e^t dt$$

$$x = e^t$$

$x^m = e^{mt}$, the integral becomes

$$x^m (\log x)^n dx = e^{mt} t^n e^t dt = e^{(m+1)t} t^n dt$$

limit change:

$$x=b \rightarrow t = \log b, \quad x=1 \Rightarrow t \cdot \log 1 = 0$$

$$I = - \int_0^{\log b} e^{(m+1)t} t^n dt$$

$$u = t^n \quad dV = e^{(m+1)t} dt$$

$$du = n t^{n-1} dt, \quad v = \frac{e^{(m+1)t}}{m+1}$$

Now,

$$\int e^{(m+1)t} t^n dt = \frac{t^n e^{(m+1)t}}{m+1} - \frac{n}{m+1} \int e^{(m+1)t} t^{n-1} dt$$

$$\int e^{(m+1)t} t^n dt = e^{(m+1)t} \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)! (m+1)^{k+1}} t^{n-k} + C$$

$$I = - \left[e^{(m+1)t} \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)! (m+1)^{k+1}} t^{n-k} \right]_{t=0}^{t=\log 6}$$

$$e^{(m+1)t} = n^{m+1} \text{ and } t = \log n$$

$$I = - \left[n^{m+1} \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)! (m+1)^{k+1}} (\log n)^{n-k} \right]_6^1$$

$$= {}_6^1 \int n^m (\log n)^n dn = \frac{(-1)^n n!}{(m+1)^{n+1}} (1^{m+1} - 6^{m+1})$$

$$\textcircled{16} \beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$$

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx, \quad \beta(m, n) = \int_0^1 t^{m-1} (1-t)^{n-1} dt$$

$$\Gamma(m) \Gamma(n) = \int_0^\infty \int_0^\infty x^{m-1} y^{n-1} e^{-(x+y)} dx dy$$

$$x = ny, \quad t = \frac{n}{n+y} \quad (\text{so } n = xt, \quad y = x(1-t))$$

$$\text{with } x \in (0, \infty), t \in (0, 1)$$

$$\left| \frac{\partial(x, y)}{\partial(x, t)} \right| = x$$

$$\Gamma(m) \Gamma(n) = \int_0^{\infty} x^{m+n-1} e^{-x} dx \int_0^1 t^{m-1} (1-t)^{n-1} dt$$

$$\Rightarrow \Gamma(m) \Gamma(n) = \Gamma(m+n) \cdot B(m, n)$$

$$\Rightarrow B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\textcircled{17} B(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$$

$$B(m, n) = \int_0^1 t^{m-1} (1-t)^{n-1} dt, \quad m, n > 0$$

$$t = \frac{x}{1+x} \quad \text{so that} \quad x = \frac{t}{1-t}, \quad dx = \frac{dt}{(1-t)^2}$$

$$\text{when } x=0 \Rightarrow t=0 \quad \text{and when } x \rightarrow \infty \Rightarrow t \rightarrow 1$$

$$B(m, n) = \int_0^1 \left(\frac{x}{1+x}\right)^{m-1} \left(\frac{1}{1+x}\right)^{n-1} \frac{dx}{(1+x)^2}$$

simplify -

$$B(m, n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$B(m, n) = B(n, m)$$

$$B(n, m) = \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

Adding the 2 integrals

$$2 B(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$$

$$B(m, n) = \frac{1}{2} \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$$

⑥ Given $\int_0^{\infty} \frac{n^{p-1}}{1+n} dn = \frac{\pi}{\sin p\pi}$. Show that $\Gamma(p)\Gamma(1-p) = \frac{\pi}{\sin p\pi}$

$n = \frac{t}{1-t}$ so $t = \frac{n}{1+n}$ $dt = \frac{dn}{(1+n)^2}$

$$\int_0^{\infty} \frac{n^{p-1}}{1+n} dn = \int_0^1 \frac{\left(\frac{t}{1-t}\right)^{p-1}}{1+\frac{t}{1-t}} \cdot \frac{dt}{(1-t)^2} = \int_0^1 t^{p-1} (1-t)^{-p} dt$$

$$B(p, 1-p) = \int_0^1 \frac{n^{p-1}}{1+n} dn$$

$B(a, b) = \frac{\Gamma(a) \cdot \Gamma(b)}{\Gamma(a+b)}$ $a=p, b=1-p$

$B(p, 1-p) = \Gamma(p) \Gamma(1-p)$

$$\int_0^{\infty} \frac{n^{p-1}}{1+n} dn = \Gamma(p) \Gamma(1-p)$$

$\Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin(\pi p)}$ $(0 < p < 1)$

$$\int_0^{\infty} \frac{n^{p-1}}{1+n} dn = \frac{\pi}{\sin(\pi p)} = \Gamma(p) \Gamma(1-p)$$

16x60
100

⑦ show that $\Gamma(n)\Gamma(n+\frac{1}{2}) = \frac{\sqrt{\pi}}{2^{2n-1}} \sqrt{2n}$ and $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$

$$\Gamma(z, z) = \int_0^1 t^{z-1} (1-t)^{z-1} dt$$

$t = \sin^2 \theta$ $dt = 2 \sin \theta \cdot d\theta \cos \theta$

$$B(z, z) = 2 \int_0^{\pi/2} \sin^{2z-1} \theta \cos^{2z-1} \theta d\theta = 2^{1-2z} \int_0^{\pi} \sin^{2z-1} (2\theta) d\theta$$

$\phi = 2\theta$

$$B(z, z) = 2^{-2z} \int_0^{\pi} \sin^{2z-1} \phi d\phi$$

$$\int_0^{\pi} \sin^m \phi d\phi = \sqrt{\pi} \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2} + \frac{1}{2})}$$

$$\Gamma(z) \Gamma(z + \frac{1}{2}) = 2^{1-2z} \sqrt{\pi} \Gamma(2z)$$

Put $z = n$

$$\Gamma(n) \Gamma(n + \frac{1}{2}) = 2^{1-2n} \sqrt{\pi} \Gamma(2n)$$

If n is +ve integer, $\Gamma(2n) = (2n-1)!$

$$\Gamma(n) \Gamma(n + \frac{1}{2}) = 2^{1-2n} \sqrt{\pi} (2n-1)!$$

Put $z = \frac{1}{4}$, Then

$$\Gamma(\frac{1}{4}) \Gamma(\frac{3}{4}) = 2^{1-1/2} \sqrt{\pi} \Gamma(\frac{1}{2}) = 2^{1/2} \sqrt{\pi} \cdot \sqrt{\pi} = \sqrt{2} \pi$$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\text{Thus } \boxed{\Gamma(\frac{1}{4}) \Gamma(\frac{3}{4}) = \pi \sqrt{2}}$$

$$\textcircled{20} \int_0^{\pi/2} \sqrt{\tan x} dx$$

$$I = \int_0^{\pi/2} (\tan x)^{1/2} dx$$

$$t = \sin^2 x \quad dt = 2 \sin x \cos x dx$$

$$dx = \frac{dt}{2\sqrt{t(1-t)}}$$

$$\tan^{1/2} x = \frac{\sin^{1/2} x}{\cos^{1/2} x} = t^{1/4} (1-t)^{-1/4}$$

$$\tan^{1/2} x = t^{1/4} (1-t)^{-1/4} \cdot \frac{dt}{2t^{1/2}(1-t)^{1/2}} = \frac{1}{2} t^{(1/4-1/2)} (1-t)^{(-1/4-1/2)} dt$$

$$= \frac{1}{2} t^{-1/4} (1-t)^{-3/4} dt$$

$$I = \frac{1}{2} \int_0^1 t^{3/4-1} (1-t)^{1/4-1} dt = \frac{1}{2} B(\frac{3}{4}, \frac{1}{4})$$

$$I = \frac{1}{2} \Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)$$

$$\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right) = \pi\sqrt{2}$$

$$\boxed{\int_0^{\frac{\pi}{2}} \sqrt{\tan u} \, du = \frac{1}{2} \pi\sqrt{2} = \frac{\pi\sqrt{2}}{2}}$$

$$\textcircled{1} \quad B(m, n) = B(m+1, n) + B(m, n+1)$$

$$B(m, n) = \int_0^1 t^{m-1} (1-t)^{n-1} dt$$

Through integral definition

$$\text{Since, } 1 = t + (1-t)$$

$$B(m, n) = \int_0^1 t^{m-1} (1-t)^{n-1} (t + (1-t)) dt = \int_0^1 \underbrace{t^m (1-t)^{n-1}}_{B(m+1, n)} dt + \int_0^1 \underbrace{t^{m-1} (1-t)^n}_{B(m, n+1)} dt$$

$$\text{Hence, } B(m, n) = B(m+1, n) + B(m, n+1)$$

From Gamma Method

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \quad \text{and} \quad \Gamma(n+1) = n \Gamma(n)$$

$$\begin{aligned} B(m+1, n) + B(m, n+1) &= \frac{\Gamma(m+1) \Gamma(n)}{\Gamma(m+1+n)} + \frac{\Gamma(m) \Gamma(n+1)}{\Gamma(m+n+1)} = \frac{m \Gamma(m) \Gamma(n) + n \Gamma(m) \Gamma(n)}{m+n \Gamma(m+n)} \\ &= \frac{(m+n) \Gamma(m) \Gamma(n)}{(m+n) \Gamma(m+n)} = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} = B(m, n) \end{aligned}$$

$$(22) \int_{-1}^1 (1-x^2)^n dx = I_n$$

Put $x = \sin \theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $dx = \cos \theta d\theta$ and $1-x^2 = \cos^2 \theta$

$$I_n = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^{2n+1} \theta d\theta = 2 \int_0^{\frac{\pi}{2}} \cos^{2n+1} \theta d\theta$$

use Beta integral, $\int_0^{\frac{\pi}{2}} \sin^{2u-1} \theta \cdot \cos^{2v-1} \theta d\theta = \frac{1}{2} B(u, v)$, Here $2u-1=0$
 $\boxed{v=n+1}$ $\boxed{u=\frac{1}{2}}$

$$I_n = 2 \cdot \frac{1}{2} B(\frac{1}{2}, n+1) = B(\frac{1}{2}, n+1) = \frac{\Gamma(\frac{1}{2}) \Gamma(n+1)}{\Gamma(n+\frac{3}{2})}$$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi}, \quad \Gamma(n+1) = n! \quad \text{and} \quad \Gamma(n+\frac{3}{2}) = \frac{(2n+1)!!}{2^{n+1}} \sqrt{\pi}$$

$$I_n = \frac{n! 2^{n+1}}{(2n+1)!!}$$

using $(2n+1)!! = \frac{(2n+1)!}{2^n n!}$ we get,

$$\boxed{\int_{-1}^1 (1-x^2)^n dx = \frac{2^{2n+1} (n!)^2}{(2n+1)!}}$$

$$\int_0^{\infty} 2^{-qn^2} dn = \frac{1}{6} \sqrt{\frac{\pi}{\ln 2}}$$

$$2^{-qn^2} = e^{-qn^2 \ln 2}$$

Set $\beta = q \ln 2 > 0$,

$$\int_a^{\infty} e^{-\beta n^2} dn = \frac{1}{2} \sqrt{\frac{\pi}{\beta}} \operatorname{erfc}(a\sqrt{\beta})$$

where $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^{\infty} e^{-t^2} dt$

$$\int_0^{\infty} 2^{-qn^2} dn = \frac{1}{2} \sqrt{\frac{\pi}{\beta}} = \frac{1}{2} \sqrt{\frac{\pi}{q \ln 2}} = \frac{1}{6} \sqrt{\frac{\pi}{\ln 2}}$$

(24) $\int_a^b (n-a)^{m-1} (b-n)^{n-1} dn = (b-a)^{m+n-1} B(m, n)$

$$I = \int_a^b (n-a)^{m-1} (b-n)^{n-1} dn$$

Set,

$$t = \frac{n-a}{b-a} \Rightarrow n = a + (b-a)t, \quad dx = (b-a)dt$$

when, $n=a \Rightarrow t=0$ and $n=b \Rightarrow t=1$

$$n-a = (b-a)t, \quad b-n = (b-a)(1-t)$$

Hence,

$$I = \int_0^1 ((b-a)t)^{m-1} ((b-a)(1-t))^{n-1} (b-a) dt$$

$$= (b-a)^{m+n-1} \int_0^1 t^{m-1} (1-t)^{n-1} dt$$

$$= (b-a)^{m+n-1} B(m, n)$$

$$(25) \int_{-\infty}^{\infty} \frac{e^{mx}}{ae^{nx}+b} dx = \frac{\pi}{n} \left(\frac{b}{a}\right)^{m/n} \left[\frac{1}{b \sin(\pi m/n)} \right]$$

$$I = \int_{-\infty}^{\infty} \frac{e^{mx}}{ae^{nx}+b} dx \quad a, b > 0$$

$t = e^x$, then $x = \ln t$, $dx = dt/t$

$$I = \int_0^{\infty} \frac{t^m}{at^n+b} \frac{dt}{t} = \int_0^{\infty} \frac{t^{m-1}}{at^n+b} dt$$

Factor b,

$$I = b^{-1} \int_0^{\infty} \frac{t^{m-1}}{1+\lambda t^n} dt, \quad \left[\lambda = \frac{a}{b} \right]$$

$$u = \lambda^{1/n} t, \quad t = \lambda^{-1/n} u, \quad dt = \lambda^{-1/n} du$$

$$t^{m-1} dt = \lambda^{-m/n} u^{m-1} du \quad \text{Hence,}$$

$$I = b^{-1} \lambda^{-m/n} \int_0^{\infty} \frac{u^{m-1}}{1+u^n} du$$

$$\int_0^{\infty} \frac{u^{m-1}}{1+u^n} du = \frac{\pi}{n \sin(\frac{\pi m}{n})}$$

$$\Rightarrow I = b^{-1} \lambda^{-m/n} \cdot \frac{\pi}{n \sin(\frac{\pi m}{n})} = \frac{\pi}{n} \frac{a^{-m/n} b^{m/n-1}}{\sin(\pi m/n)}$$

$$\boxed{\int_{-\infty}^{\infty} \frac{e^{mx}}{ae^{nx}+b} dx = \frac{\pi}{n} \frac{1}{b} \left(\frac{b}{a}\right)^{m/n} \frac{1}{\sin(\pi m/n)}}, \quad 0 < m < n, \quad a, b > 0$$

$$\sqrt{\frac{1+i}{1-i}} \times \sqrt{\frac{1+i}{1+i}} = \sqrt{\frac{(1+i)^2}{1-i^2}} = \sqrt{\frac{1+2i+i^2}{1-i^2}} = \sqrt{\frac{2i}{2}} = \sqrt{i}$$

gn Polar Form: $i = e^{i\pi/2}$

$$\sqrt{i} = e^{i\pi/4}$$

Modulus = 1

Principle Argument = $\pi/4$

$$2 \cdot (4+2i)(-3+\sqrt{2}i)$$

$$= -12 + 4\sqrt{2}i - 6i - 2\sqrt{2}$$

$$= (-12 + \sqrt{2}i) + (-6i + 2i^2\sqrt{2})$$

$$= (-12 - 2\sqrt{2}) + i(4\sqrt{2} - 6)$$

$$r = \sqrt{(-12-2\sqrt{2})^2 + (4\sqrt{2}-6)^2}$$

$$(-12-2\sqrt{2})^2 = 144 + 48\sqrt{2} + 8 = 152 + 48\sqrt{2}$$

$$(4\sqrt{2}-6)^2 = 32 - 48\sqrt{2} + 36 = 68 - 48\sqrt{2}$$

$$r^2 = 220$$

$$r = \sqrt{220} = 2\sqrt{55} \text{ (modulus)}$$

$$\tan \theta = \frac{\text{Imag.}}{\text{Real}} = \frac{4\sqrt{2}-6}{-12-2\sqrt{2}} \quad (\text{Multiply Num \& Den. by } -1)$$

$$\tan \theta = \frac{6-4\sqrt{2}}{12+2\sqrt{2}} = \frac{3-2\sqrt{2}}{6+\sqrt{2}}$$

$$\text{Argument } \theta = \pi + \tan^{-1}\left(\frac{4\sqrt{2}-6}{-12-2\sqrt{2}}\right)$$

$$(27) \quad z^4 + 1 = 0$$

$z^4 = -1$ we know that,
 $-1 = e^{i\pi} = e^{i(\pi + 2k\pi)}$, $k \in \mathbb{Z}$

$$z = e^{i(\pi + 2k\pi)/4} = e^{i(\pi/4 + k\pi/2)}, \quad k = 0, 1, 2, 3$$

k	Angle (in radian)	$z = \cos \theta + i \sin \theta$
0	$\pi/4$	$\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$
1	$3\pi/4$	$-\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$
2	$5\pi/4$	$-\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$
3	$7\pi/4$	$\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$

$$z = e^{i\pi/4}, e^{i3\pi/4}, e^{i5\pi/4}, e^{i7\pi/4}$$

(28) $\sin(\alpha + i\beta) = x + iy$, prove that

(a) $\frac{x^2}{\cosh^2 \beta} + \frac{y^2}{\sinh^2 \beta} = 1$; (b) $\frac{x^2}{\sin^2 \alpha} - \frac{y^2}{\cos^2 \alpha} = 1$

$$\sin(\alpha + i\beta) = \sin \alpha \cosh \beta + i \cos \alpha \sinh \beta$$

$$x = \sin \alpha \cosh \beta \quad \text{and} \quad y = \cos \alpha \sinh \beta$$

Divide x by $\cosh \beta$ and y by $\sinh \beta$

$$\frac{x^2}{\cosh^2 \beta} = \sin^2 \alpha, \quad \frac{y^2}{\sinh^2 \beta} = \cos^2 \alpha$$

Adding these

$$\frac{x^2}{\cosh^2 \beta} + \frac{y^2}{\sinh^2 \beta} = \sin^2 \alpha + \cos^2 \alpha = 1$$

divide x by $\sinh^2 \alpha$ and y by $\cosh^2 \alpha$

$$\frac{x^2}{\sinh^2 \alpha} = \cosh^2 \beta, \quad \frac{y^2}{\cosh^2 \alpha} = \sinh^2 \beta$$

Subtract second from first

$$\frac{x^2}{\sinh^2 \alpha} - \frac{y^2}{\cosh^2 \alpha} = \cosh^2 \beta - \sinh^2 \beta = 1 \quad \text{--- (b) part}$$

29) $\sin z = 2$ Find all values of z

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\frac{e^{iz} - e^{-iz}}{2i} = 2$$

$$e^{iz} - e^{-iz} = 4i$$

$$\text{Let } w = e^{iz}$$

$$e^{-iz} = 1/w$$

$$w - \frac{1}{w} = 4i$$

$$w^2 - 4iw - 1 = 0$$

$$w = \frac{4i \pm \sqrt{(4i)^2 + 4}}{2} = \frac{4i \pm \sqrt{-16 + 4}}{2} = \frac{4i \pm \sqrt{-12}}{2} = \frac{4i \pm 2i\sqrt{3}}{2} = i(2 \pm \sqrt{3})$$

$$w = i(2 + \sqrt{3}) \text{ or } w = i(2 - \sqrt{3})$$

$$w = e^{iz}$$

$$\Rightarrow e^{iz} = i(2 \pm \sqrt{3})$$

$$iz = \ln(i(2 \pm \sqrt{3})) + 2n\pi i, \quad n \in \mathbb{Z}$$

for $i(2 + \sqrt{3})$

$$\text{Modulus} = |i(2 + \sqrt{3})| = 2 + \sqrt{3}$$

$$\text{Argument} = \pi/2$$

$$\ln[i(2+\sqrt{3})] = \ln(2+\sqrt{3}) + i\pi/2$$

For $i(2-\sqrt{3})$

Modulus $r_2 = 2-\sqrt{3}$

Argument = $\pi/2$

$$\ln[i(2-\sqrt{3})] = \ln(2-\sqrt{3}) + i\pi/2$$

$$iz = \ln[i(2+\sqrt{3})] + 2n\pi i$$

divide both side by i

$$z = -i[\ln(2+\sqrt{3}) + i\pi/2] + 2n\pi = \frac{\pi}{2} + 2n\pi - i\ln(2+\sqrt{3})$$

$$z = \frac{\pi}{2} + 2n\pi + i\ln(2+\sqrt{3})$$

$$\textcircled{50} \log(1+i) \log i$$

$$\log i = i\pi/2 \quad [\text{since } i = e^{i\pi/2}]$$

$$(1+i) \log i = (1+i) \left(i\frac{\pi}{2} \right) = \frac{\pi i}{2} (1+i)$$

$$= \frac{\pi}{2} (i+ i^2) = \frac{\pi}{2} (i-1) = \frac{\pi}{2} (-1+i)$$

$$\text{In Polar form: } r = \sqrt{(-1)^2 + 1^2} = \sqrt{2} \quad ; \quad \theta = 3\pi/4$$

$$(1+i) \log i = \frac{\pi}{2} \sqrt{2} e^{i3\pi/4}$$

Taking log

$$\log((1+i) \log i) = \log\left(\frac{\pi}{2} \sqrt{2}\right) + i\frac{3\pi}{4}$$

$$\text{Real Part: } \ln\left(\frac{\pi}{2} \sqrt{2}\right) = \ln \pi - \ln 2 + \frac{1}{2} \ln 2 = \ln \pi - \frac{1}{2} \ln 2$$

$$\text{Imaginary Part} = 3\pi/4$$

Q1) (a) $(i)^i$

$$i = e^{i\pi/2 + 2n\pi i}, n \in \mathbb{Z}$$

So,

$$(i)^i = [e^{i(\pi/2 + 2n\pi)}]^i = e^{-(\pi/2 + 2n\pi)}$$

General value: $(i)^i = e^{-(\pi/2 + 2n\pi)}, n \in \mathbb{Z}$

Principal value (for $n=0$): $(i)^i = e^{-\pi/2}$

(b) $(1 + \sqrt{3}i)^{i+1}$

$$1 + \sqrt{3}i = 2(\cos \pi/3 + i \sin \pi/3) = 2e^{i\pi/3}$$

$$(1 + \sqrt{3}i)^{i+1} = (2e^{i\pi/3})^{i+1} = 2^{i+1} e^{i(i+1)\pi/3}$$

$$\text{Simplifying} = 2^{i+1} e^{i^2\pi/3} e^{i\pi/3} = 2^{i+1} e^{-\pi/3} e^{i\pi/3}$$

$$(1 + \sqrt{3}i)^{i+1} = 2e^{-\pi/3} e^{i(\pi/3 + \ln 2)}$$

$$\text{Principal value: } (1 + \sqrt{3}i)^{i+1} = 2e^{-\pi/3} [\cos(\pi/3 + \ln 2) + i \sin(\pi/3 + \ln 2)]$$

Q2) $\tan^{-1}z$ in u+iv form

Let,

$$\tan^{-1}z = \frac{i}{2} \log \left(\frac{1-iz}{1+iz} \right) \quad z = x+iy$$

$$\tan^{-1}(x+iy) = \frac{1}{2} \tan^{-1} \left(\frac{2x}{1-x^2-y^2} \right) + \frac{i}{4} \log \left(\frac{x^2 + (y+1)^2}{x^2 + (y-1)^2} \right)$$

So,

$$u = \frac{1}{2} \tan^{-1} \left(\frac{2x}{1-x^2-y^2} \right) \quad v = \frac{1}{4} \log \left(\frac{x^2 + (y+1)^2}{x^2 + (y-1)^2} \right)$$

$$\text{Q3) (a) } \tan^{-1}z = \frac{-i}{2} \log \left(\frac{1+iz}{1-iz} \right) = \frac{1}{2} \log \left(\frac{i+iz}{i-iz} \right); \quad z \neq \pm i$$

$$\tan w = z \Rightarrow e^{2iw} = \frac{1+iz}{1-iz}$$

Taking log,

$$2iw = \log \left(\frac{1+iz}{1-iz} \right)$$

Hence

$$w = \frac{i}{2} \log \left(\frac{1-iz}{1+iz} \right) = -\frac{i}{2} \log \left(\frac{1+iz}{1-iz} \right)$$

or equivalently,

$$\tan^{-1} z = \frac{i}{2} \log \left(\frac{1+z}{1-z} \right)$$

$$(b) \cot h^{-1} z = \frac{1}{2} \log \left| \frac{z+1}{z-1} \right| \quad iz \neq \pm 1$$

$$\text{Let } \cot h w = z \Rightarrow e^{2w} = \frac{z+1}{z-1}$$

Then, Taking log

$$2w = \log \left(\frac{z+1}{z-1} \right)$$

Hence,

$$\cot h^{-1} z = \frac{1}{2} \log \left(\frac{z+1}{z-1} \right)$$

$$(34) \lim_{z \rightarrow 0} \frac{z}{|z|} \text{ does not exist}$$

$$z = x+iy, \text{ where } x, y \in \mathbb{R}$$

Then

$$|z| = \sqrt{x^2 + y^2}$$

So:

$$\frac{z}{|z|} = \frac{x+iy}{\sqrt{x^2+y^2}} = \frac{x}{\sqrt{x^2+y^2}} + i \frac{y}{\sqrt{x^2+y^2}}$$

$$(1) y = 0$$

Then $z = x$ and as $x \rightarrow 0$

$$\frac{z}{|z|} = \frac{x}{|x|} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Inconsistency

$$(2) x = 0$$

$$z = iy \quad |z| = |y|$$

So:

$$\frac{z}{|z|} = \frac{iy}{|y|} = \begin{cases} i & y > 0 \\ -i & y < 0 \end{cases}$$

The limit $\lim_{z \rightarrow 0} \frac{z}{|z|}$ does not exist

(35) $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ does not exist

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$\frac{z}{\bar{z}} = \frac{x + iy}{x - iy}$$

$$\frac{z}{\bar{z}} = \frac{(x + iy)(x + iy)}{(x - iy)(x + iy)} = \frac{(x + iy)^2}{x^2 + y^2}$$

$$(x + iy)^2 = x^2 + 2ixy + y^2$$

$$\frac{z}{\bar{z}} = \frac{x^2 - y^2}{x^2 + y^2} + i \frac{2xy}{x^2 + y^2}$$

① Along x ($y=0$)

$$\frac{z}{\bar{z}} = \frac{x^2 - 0}{x^2 + 0} + i \frac{0}{x^2} = 1$$

② Along y ($x=0$)

$$\frac{z}{\bar{z}} = \frac{0 - y^2}{0 + y^2} + i \frac{0}{y^2} = -1$$

③ Along $y=x$

$$\frac{z}{\bar{z}} = \frac{x^2 - x^2}{x^2 + x^2} + i \frac{2x^2}{2x^2} = 0 + i = i$$

Hence, $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ does not exist

(36) Assume $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at $z_0 = (x_0, y_0)$

Then

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

1. Take $h = \Delta x \in \mathbb{R}$ Then

$$\frac{f(z_0 + \Delta x) - f(z_0)}{\Delta x} = \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x}$$

$$f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0)$$

2. Take $h = i\Delta y$ (pure imaginary):

$$\frac{f(z_0 + i\Delta y) - f(z_0)}{i\Delta y} = \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{i\Delta y} + \frac{i v(x_0, y_0 + \Delta y) - i v(x_0, y_0)}{i\Delta y}$$

$$f'(z_0) = -i u_y(x_0, y_0) + v_y(x_0, y_0)$$

$$u_x + i v_x = -i u_y + v_y$$

$$u_x = v_y \quad v_x = -u_y$$

or equivalently

$$\boxed{u_x = v_y \quad u_y = -v_x}$$

$$(87) f(z) = u(x, y) + i v(x, y)$$

$$z_0 = x_0 + i y_0$$

$$u_x = v_y, \quad u_y = -v_x$$

$$f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0) = v_y(x_0, y_0) - i u_y(x_0, y_0)$$

$$\Delta u = u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) = u_x(x_0, y_0) \Delta x + u_y(x_0, y_0) \Delta y + o(r)$$

$$\Delta v = v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0) = v_x(x_0, y_0) \Delta x + v_y(x_0, y_0) \Delta y + o(r)$$

$$r = \sqrt{(\Delta x)^2 + (\Delta y)^2} \quad \text{and} \quad o(r)/r \rightarrow 0 \quad \text{as} \quad r \rightarrow 0$$

so

$$\begin{aligned} \Delta f &= \Delta u + i \Delta v \\ &= (u_x \Delta x + u_y \Delta y) + i (v_x \Delta x + v_y \Delta y) + o(r) \\ &= (u_x + i v_x) \Delta x + (u_y + i v_y) \Delta y + o(r) \end{aligned}$$

$$u_y = -v_x, \quad v_y = u_x$$

$$\begin{aligned} \Delta f &= (u_x + i v_x) \Delta x + (-v_x + i u_x) \Delta y + o(r) \\ &= (u_x + i v_x) (\Delta x + i \Delta y) + o(r) \\ &= (u_x + i v_x) \Delta z + o(r) \end{aligned}$$

Divide by Δz

$$\frac{\Delta f}{\Delta z} = u_x + i v_x + \frac{o(r)}{\Delta z}$$

$$\text{But } |\Delta z| = r$$

$$f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0)$$

Using Cauchy - Riemann relation

$$u_x + i v_x = v_y - i u_y$$

or

$$f'(z_0) = v_y(x_0, y_0) - i u_y(x_0, y_0)$$

Q9) $z = re^{i\theta}$ and $u = r \cos \theta$ $v = r \sin \theta$

$$f(z) = u(x,y) + i v(x,y)$$

Step 1: chain Rule relations

$$u_x = u_r \cos \theta - v_r \sin \theta \quad v_x = v_r \cos \theta + u_r \sin \theta$$

$$u_\theta = -r u_r \sin \theta + r v_r \cos \theta \quad v_\theta = -r v_r \sin \theta + r u_r \cos \theta$$

Solve the linear system we get

$$u_x = \cos \theta u_r - \frac{\sin \theta}{r} v_\theta$$

$$v_x = \cos \theta v_r - \frac{\sin \theta}{r} u_\theta$$

$$u_y = \sin \theta u_r + \frac{\cos \theta}{r} v_\theta$$

$$v_y = \sin \theta v_r + \frac{\cos \theta}{r} u_\theta$$

Step 2: substitute the cartesian C-R equation

$$u_x = v_y \quad u_y = -v_x$$

$$\cos \theta u_r - \frac{\sin \theta}{r} v_\theta = \sin \theta v_r + \frac{\cos \theta}{r} u_\theta$$

$$\sin \theta u_r + \frac{\cos \theta}{r} v_\theta = -\cos \theta v_r + \frac{\sin \theta}{r} u_\theta$$

$$u_\theta = \frac{1}{r} v_\theta$$

Polar Cauchy-Riemann Eq.

$$u_r = \frac{1}{r} v_\theta \quad v_r = -\frac{1}{r} u_\theta$$

$$f'(z) = u_x + i v_x = u_x + i v_x$$

The net polar expression is:

$$f'(z) = e^{i\theta} (u_x + i v_x) = \frac{e^{-i\theta}}{r} (v_\theta - i u_\theta)$$

$$f(z) = \begin{cases} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2} & , z \neq 0 \\ 0 & , z = 0 \end{cases}$$

$$f(z) = 4 + i$$

$$f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2} = \frac{(x^3-y^3) + i(x^3+y^3)}{x^2+y^2}$$

$$u(x,y) = \frac{x^3-y^3}{x^2+y^2}$$

$$v(x,y) = \frac{x^3+y^3}{x^2+y^2}$$

$$r = \sqrt{x^2+y^2}$$

$$|f(z)| \sim \frac{r^3}{r^2} = r \rightarrow 0 \text{ as } r \rightarrow 0$$

Therefore f is continuous at $z=0$

$$f'(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z \rightarrow 0} \frac{f(z)}{z}$$

$$z = re^{i\theta} = x + iy = r(\cos\theta + i\sin\theta)$$

$$\text{Then } x = r\cos\theta, y = r\sin\theta$$

$$f(z) = \frac{r^3(\cos^3\theta - \sin^3\theta) + i r^3(\cos^3\theta + \sin^3\theta)}{r^2} = r[(\cos^3\theta - \sin^3\theta) + i(\cos^3\theta + \sin^3\theta)]$$

Thus

$$\frac{f(z)}{z} = \frac{r[(\cos^3\theta - \sin^3\theta) + i(\cos^3\theta + \sin^3\theta)]}{r(\cos\theta + i\sin\theta)} = \frac{(\cos^3\theta - \sin^3\theta) + i(\cos^3\theta + \sin^3\theta)}{\cos\theta + i\sin\theta}$$

$$(a) y=0 \Rightarrow \theta=0$$

$$f(z) = \frac{x^3(1+i)}{x^2} = x(1+i) \Rightarrow \frac{f(z)}{z} = \frac{x(1+i)}{x} = 1+i$$

$$(b) x=0 \Rightarrow \theta=\pi/2$$

$$f(z) = \frac{-y^3(1-i)}{y^2} = -y(1-i) \Rightarrow \frac{f(z)}{z} = \frac{-y(1-i)}{iy} = i(1-i) = i+1$$

$$(c) y=x, \text{ limit is } 1+i/2$$

but along x axis it is $1+i$

Hence, the limit depends upon direction

The function $f(z)$ is not differentiable at $z=0$ so it is not analytic
There

40) a) $f(z) = -x^2 + ixy + iy^2 + n - 2ay + i(bx - cy)$

$u(x,y) = x - 2ay, \quad v(x,y) = bx - cy$

$u_x = v_y, \quad u_y = -v_x$

$u_x = 1, \quad u_y = -2a, \quad v_x = b, \quad v_y = -c$

so $1 = -c \Rightarrow c = -1, \quad -2a = -b \Rightarrow b = 2a$

Thus the family of solution is

$b = 2a, \quad c = -1$ (with a arbitrary)

b) $u_x = v_y: -2x + y = bx + 2cy$

$f(z) = u + iv \quad u(x,y) = -x^2 + xy + y^2 + 1 \quad v(x,y) = ax^2 + bxy + cy^2$

$u_x = v_y, \quad u_y = -v_x$

$u_x = -2x + y, \quad u_y = x + 2y$

$v_x = 2ax + by, \quad v_y = b + 2cy$

$u_x = v_y: -2x + y = b + 2cy$

$-2 = b, \quad 1 = 2c \rightarrow b = -2, c = 1/2$

From $u_y = -v_x: x + 2y = -(2ax + by)$

$1 = -2a, \quad 2 = -b \Rightarrow a = -1/2, b = -2$

so $a = -\frac{1}{2}, b = -2, c = \frac{1}{2}$

$f(z) = u + iv = 1 + (-1 - i/2)z^2$

$$\textcircled{11} \quad U+V = (x+y)(2-4xy+x^2+y^2)$$

$$g(x,y) = u+v \quad P = u+iv$$

$$u_x = v_y = g_y - u_y, \quad u_y = -v_x = -(g_x - u_x)$$

So

$$u_x + u_y = g_y \quad -u_x + u_y = -g_x$$

$$u_x = \frac{1}{2}(g_x + g_y), \quad u_y = \frac{1}{2}(g_y - g_x)$$

compute g (expand)

$$g = (x+y)(2-4xy+x^2+y^2) = 2(x+y) + x^3 + y^3 - 3x^2y - 3xy^2$$

Hence

$$g_x = 2 + 3x^2 - 6xy - 3y^2, \quad g_y = 2 + 3y^2 - 3x^2 - 6xy$$

Thus

$$u_x = 2 - 6xy \quad u_y = 3(y^2 - x^2)$$

Integrate on wrt x :

$$u(x,y) = \int (2 - 6xy) dx = 2x - 3x^2y + h(y)$$

Diff. wrt y and equate to the expression for u_y

$$u_y = -3x^2 + h'(y) = 3(y^2 - x^2) \Rightarrow h'(y) = 3y^2 \Rightarrow h(y) = y^3 + C$$

Take $C=0$

$$\boxed{u(x,y) = 2x - 3x^2y + y^3}$$

$$\boxed{v(x,y) = 2y + x^3 - 3xy^2}$$

Express P in terms of $z = x+iy$

$$iz^3 = i(x^3 + 3ix^2y - 3xy^2 - iy^3) = -3x^2y + y^3 + i(x^3 - 3xy^2)$$

So

$$2z + iz^3 = (2x - 3x^2y + y^3) + i(2y + x^3 - 3xy^2) = u+iv$$

Analytic Function is:

$$\boxed{f(z) = 2z + iz^3}$$

(12) $f(z) = u + iv$ is analytic function and

$$u - v = e^x [(x-y) \sin y - (x+y) \cos y]$$

Step 1: Let $g(x, y) = u - v$

$$u_x = v_y, \quad u_y = -v_x$$

Now diff. $g = u - v$

$$g_x = u_x - v_x \quad g_y = v_y - v_y$$

Using C-R equations:

$$u_x = v_y \Rightarrow v_y = u_x$$

so

$$g_x = u_x + u_x, \quad g_y = u_y - u_x$$

Solve for u_x, u_y

$$g_x = u_x + u_x$$

$$g_y = u_y - u_x$$

Add and subtract to get

$$u_x = \frac{1}{2} (g_x - g_y) \quad u_y = \frac{1}{2} (g_x + g_y)$$

Step 3: Compute g_x, g_y

$$g = e^x [(x-y) \sin y - (x+y) \cos y]$$

$$g'_x = e^x [(x-y+1) \sin y - (x+y+1) \cos y]$$

diff. wrt y

$$g_x = e^x [-(x-y) \cos y - \sin y - (x+y) \sin y - \cos y]$$

$$g_y = e^x [-(x-y+1) \cos y - (x+y+1) \sin y]$$

$$u_x = \frac{1}{2} e^x [g_x - g_y]$$

$$= \frac{1}{2} e^x [(x-y+1) \sin y - (x+y+1) \cos y - (- (x-y+1) \cos y - (x+y+1) \sin y)]$$

$$u_x = \frac{1}{2} e^x [(x-y+1) + x+y+1) (\sin y + \cos y)]$$

$$u_x = e^x(x+1)(\sin y + \cos y)$$

Similarly,

$$v_y = \frac{1}{2} e^x(g_x + g_y)$$

$$v_y = -e^x(y+1)(\sin y + \cos y)$$

Step 5: integrate to get u

$$u = e^x(x)(\sin y + \cos y) + \phi(y)$$

diff. wrt y

$$u_y = e^x(x)(\cos y - \sin y) + \phi'(y)$$

equating

$$e^x(x)(\cos y - \sin y) + \phi'(y) = -e^x(y+1)(\sin y + \cos y)$$

$$f(z) = e^x[(x-y+1) + i(x+y+1)](\sin y + i \cos y)$$

$$u = e^x[(x+1)\cos y - y\sin y]$$

$$v = e^x[(x+1)\sin y + y\cos y]$$

$$f(z) = e^z(z+1)$$

43) $f(z) = u(x,y) + iv(x,y)$

$$u = \left(n + \frac{1}{2}\right) \cos \theta \quad v = \left(n - \frac{1}{2}\right) \sin \theta$$

$$n = r \cos \theta \quad \text{and } \theta = \arg z$$

$$u(r, \theta) = r \cos^2 \theta + \frac{1}{2} \cos \theta, \quad v(r, \theta) = r \cos \theta \sin \theta - \frac{1}{2} \sin \theta \quad [\text{in Polar form}]$$

$$u_r = \cos^2 \theta \quad v_r = \cos \theta \sin \theta$$

$$u_\theta = \frac{1}{r} v \quad v_\theta = -\frac{1}{r} u$$

Compute the θ -partials:

$$u_\theta = r \cdot 2 \cos \theta (-\sin \theta) + \frac{1}{2} (-\sin \theta) = -r \sin 2\theta - \frac{1}{2} \sin \theta$$

$$v_\theta = r (\cos^2 \theta - \sin^2 \theta) - \frac{1}{2} \cos \theta = r \cos 2\theta - \frac{1}{2} \cos \theta$$

Then

$$\frac{1}{r} v_\theta = \cos 2\theta - \frac{1}{2r} \cos \theta$$

$$f'(z) = e^{-i\theta} (\cos^2 \theta + i \cos \theta \sin \theta) = e^{i\theta} \cos \theta (\cos \theta + i \sin \theta) = \cos \theta$$

So the exp. $\cos \theta$ is what the formal polar derivative formula yields, but since the CR eqs do not hold,

The function is not analytic

44) Let $f = u + iv$ be analytic on a connected domain D and assume $|f(z)|$ is constant on D .

$$|f|^2 = u^2 + v^2 = C \quad (\text{constant})$$

Diff. wrt x and y

$$(u^2 + v^2)_x = 2(u u_x + v v_x) = 0, \quad (u^2 + v^2)_y = 2(u u_y + v v_y) = 0$$

$$u u_x + v v_x = 0$$

$$u u_y + v v_y = 0$$

becomes

$$u v_y + v u_x = 0$$

$$-u v_x + v v_y = 0$$

The linear system is written as

$$\begin{pmatrix} v & u \\ -u & v \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Hence, u and v are const, so f is constant on D .

45) $u(x,y) = 2x + y^3 - 3x^2y$

Harmonicity

compute second partials

$$u_x = 2 - 6xy, \quad u_{xx} = -6y$$

$$u_y = 3y^2 - 3x, \quad u_{yy} = 6y$$

$$u_{xx} + u_{yy} = -6y + 6y = 0$$

2. conjugate harmonic $v(x,y)$ -

$$v(x,y) = \int (2 - 6xy) dy = 2y - 3xy^2 + h(x)$$

Diff in x : $v_x = -3y^2 + h'(x)$

From CR: $v_x = -u_y = -(-3y^2 - 3x) = 3y^2 + 3x$

Thus $h'(x) = 3x$, so $h(x) = \frac{3}{2}x^2 + C$

$$v(x,y) = \frac{3}{2}x^2 + 2y - 3xy^2 + C$$

we take $C=0$ for simplicity

3. $z = x + iy$

$$f(z) = u + iv = (2x + y^3 - 3x^2y) + i(\frac{3}{2}x^2 + 2y - 3xy^2)$$

$$f(z) = 2z + iz^3 + ic$$

with $c=0$

$$\boxed{f(z) = 2z + iz^3}$$

46) Take $v(x,y) = e^x \sin y$

Harmonicity

$$v_x = e^x \sin y, \quad v_{xx} = e^x \sin y$$

$$v_y = e^x \cos y, \quad v_{yy} = -e^x \sin y$$

so, $v_{xx} + v_{yy} = e^x \sin y - e^x \sin y = 0$

$$u_x = v_y = e^x \cos y, \quad u_y = -v_x = -e^x \sin y$$

Integrate u_x

$$u(x,y) = \int e^x \cos y dx = e^x \cos y + H(y)$$

$$u(x,y) = e^x \cos y$$

$$f(z) = u + iv = e^x (\cos y + i \sin y) = e^{x+iy} = e^z$$

47) $u(x,y) = x^2 \cos 2y$

$$v(x,y) = x^2 \sin 2y = x^2 y^2$$

Harmonicity

$$u_x = 2x \quad u_{xx} = 2$$

$$u_y = -2y \quad u_{yy} = -2$$

$$u_{xx} + u_{yy} = 0, \text{ Hence, } u \text{ is harmonic}$$

$$v_y = u_x = 2x \quad v_x = -u_y = 2y$$

Integrating

$$v(x,y) = 2xy + C$$

In polar form $v = x^2 \sin 2\theta$

$$v(x,y) = 2xy \quad [C=0 \text{ for simplicity}]$$

Corresponding analytic function

$$f(z) = u + iv = (x^2 - y^2) + i(2xy) = (x + iy)^2 = z^2$$

$$\boxed{f(z) = z^2}$$

48) Let $f = u + iv$ be analytic in a connected domain D and set $M = |f|$

1. If f has 0 in D , then $M \geq 0$ is harmonic and attains its minimum 0 inside D , so by minimum principle $M \equiv 0$ and hence $f \equiv 0$

2. If f has no zeros in D , then $M > 0$. Since $\Delta M = 0$

$$\Delta (\ln M) = \frac{M \Delta M - |\nabla M|^2}{M^2} = -\frac{|\nabla M|^2}{M^2} \leq 0$$

But $\ln M$ is harmonic, so $\Delta (\ln M) = 0$. Hence $|\nabla M|^2 = 0$, so M is constant. An analytic function with constant modulus is constant

Therefore a nonconstant analytic function can't have $|f(z)|$ harmonic

49) If $f = U + iV$ is analytic then U and V are harmonic, so

$$\Delta U = U_{xx} + U_{yy} = 0 \quad \Delta V = V_{xx} + V_{yy} = 0$$

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$$

Thus for U and V we obtain

$$\frac{\partial^2 U}{\partial r^2} + \frac{1}{r} \frac{\partial U}{\partial r} + \frac{1}{r^2} \frac{\partial^2 U}{\partial \theta^2} = 0$$

$$\frac{\partial^2 V}{\partial r^2} + \frac{1}{r} \frac{\partial V}{\partial r} + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} = 0$$

$$U_r = \frac{1}{r} V_\theta \quad V_r = -\frac{1}{r} U_\theta$$

$$U_{rr} = \frac{\partial}{\partial r} \left(\frac{1}{r} V_\theta \right) = -\frac{1}{r^2} V_\theta + \frac{1}{r} V_{r\theta}$$

$$V_{r\theta} = \frac{\partial}{\partial \theta} \left(-\frac{1}{r} U_\theta \right) = -\frac{1}{r} U_{\theta\theta}$$

substitute this $V_{r\theta}$ into the expression for U_{rr} .

$$U_{rr} = -\frac{1}{r^2} V_\theta + \frac{1}{r} \left(-\frac{1}{r} U_{\theta\theta} \right) = -\frac{1}{r^2} V_\theta - \frac{1}{r^2} U_{\theta\theta}$$

But from CR relation

$$U_{rr} + \frac{1}{r} U_r + \frac{1}{r^2} U_{\theta\theta} = \left(-\frac{1}{r^2} V_\theta - \frac{1}{r^2} U_{\theta\theta} \right) + \frac{1}{r^2} V_\theta + \frac{1}{r^2} U_{\theta\theta} = 0$$

Thus

$$U_{rr} + \frac{1}{r} U_r + \frac{1}{r^2} U_{\theta\theta} = 0$$

$$V_{rr} + \frac{1}{r} V_r + \frac{1}{r^2} V_{\theta\theta} = 0$$

So $f = u + iv$ with real valued $u(x, y)$ and $v(x, y)$

$$|f| = \sqrt{u^2 + v^2}, \quad \frac{\partial |f|}{\partial x} = \frac{u u_x + v v_x}{|f|}, \quad \frac{\partial |f|}{\partial y} = \frac{u u_y + v v_y}{|f|}$$

so

$$\left(\frac{\partial |f|}{\partial x} \right)^2 + \left(\frac{\partial |f|}{\partial y} \right)^2 = \frac{(u u_x + v v_x)^2 + (u u_y + v v_y)^2}{u^2 + v^2}$$

substitute $u_y = -v_x$, $v_y = u_x$ into numerator and use algebraic identity

$$(ap + bq)^2 + (aq - bp)^2 = (a^2 + b^2)(p^2 + q^2)$$

$$a = u, b = v, p = u_x, q = v_x$$

$$(u u_x + v v_x)^2 + (u u_y + v v_y)^2 = (u^2 + v^2)(u_x^2 + v_x^2)$$

Hence

$$\left(\frac{\partial |f|}{\partial x} \right)^2 + \left(\frac{\partial |f|}{\partial y} \right)^2 = \frac{(u^2 + v^2)(u_x^2 + v_x^2)}{u^2 + v^2} = u_x^2 + v_x^2$$

But $f'(z) = u_x + i v_x$; so $|f'(z)|^2 = u_x^2 + v_x^2$ Therefore

$$\left(\frac{\partial |f|}{\partial x} \right)^2 + \left(\frac{\partial |f|}{\partial y} \right)^2 = |f'(z)|^2$$